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2.4 Number System Conversions

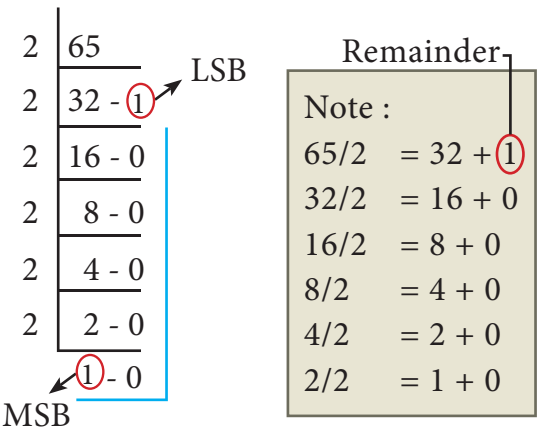
2.4.1 Decimal to Binary Conversion

Generally two methods followed.

Method 1: To convert Decimal to Binary "**Repeated Division by 2**" method can be used. Any Decimal number divided by 2 will leave a remainder of 0 or 1. Repeated division by 2 will leave a sequence of 0s and 1s that become the binary equivalent of the decimal number. Suppose it is required to convert the decimal number N into binary form, dividing N by 2 in the decimal system, we will obtain a quotient N1 and a remainder R1, where R1 can have a value of either 0 or 1. The process is repeated until the quotient becomes 0 or 1. When the quotient is '0' or '1', it is the final remainder value. Write the final answer starting from final remainder value obtained to the first remainder value obtained.

Example

Convert $(65)_{10}$ into its equivalent binary number



$(65)_{10} = (1\ 0\ 0\ 0\ 0\ 0\ 1)_2$

Method 2 : Sum of Powers of 2.

A decimal number can be converted into a binary number by adding up the powers of 2 and then adding bits as needed to obtain the total value of the number.

a) Find the largest power of 2 that is smaller than or equal to 65.

$$65_{10} > 64_{10}$$

b) Set the 64's bit to 1 and subtract 64 from the original number

$$65\text{-}64=1$$

c) 32 is greater than the remaining total. Therefore, set the 32's bit to 0.

d) 16 is greater than the remaining total. Therefore, set the 16's bit to 0.

e) 8 is greater than the remaining total. Therefore, set the 8's bit to 0.

f) 4 is greater than the remaining total. Therefore, set the 4's bit to 0.

g) 2 is greater than the remaining total. Therefore, set the 2's bit to 0.

h) As the remaining value is equivalent to 1's bit, set it to 1.

$$1\text{-}1=0$$

Conversion is complete $65_{10} = (1000001)_2$

Example

The conversion steps can be given as follows:

Given Number : 65